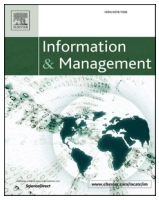
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Within- and cross-group spillover effects in influencer marketing: The heterogeneity between micro- and macro-influencers

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ABSTRACT

The meteoric growth of social media has driven firms to leverage influencers in their advertising campaigns, with a preference for those possessing a larger follower base. However, existing research has predominantly examined the role of influencer popularity in isolation, neglecting the interdependencies among influencers within a firm's campaign. In contrast, this study categorizes influencers into two groups—macro- and micro-influencers—based on their popularity, and examines the effect of within- and cross-group spillovers on influencers' conversion performance. Leveraging data from a real marketing campaign comprising 163 promotional videos of 111 influencers over a 38-month period and utilizing a fixed-effects panel data regression, we found that both within- and cross-group spillovers are significant solely among micro-influencers. Based on these findings, we discuss the theoretical implications for advertising planning and managerial implications for marketing managers seeking to optimize their influencer marketing expenditure.

1. Introduction

Influencer marketing has emerged as a pivotal strategy for firms seeking to leverage the persuasive power of social media influencers. These influencers, whose follower counts vary significantly, are categorized as macro-influencers with large followings and micro-influencers with smaller yet highly engaged audiences. Firms are increasingly allocating their advertising budgets to influencer marketing to generate positive word-of-mouth (WOM) regarding their products [1]. For instance, a recent report indicates that 85.8 percent of surveyed firms would allocate a budget to influencer marketing, and 59.4 percent of respondents intend to increase their influencer marketing budgets by 2024 [2].

As firms extensively integrate influencers into their advertising strategies, consumers are inevitably exposed to repeated advertisements (ads) from multiple influencers. This underscores the importance of examining the effectiveness of influencer marketing [3], particularly concerning influencer popularity. The significant variation in influencers' follower counts creates substantial cost discrepancies among influencers with differing popularity levels [4,5]. Therefore, understanding how to leverage the interdependencies among influencers with varying popularity levels is essential.

This inquiry intersects with both the spillover effect and influencer literature. On the one hand, previous studies on spillover effects have examined their benefits across various contexts. For instance, Krijestorac, Garg, and Mahajan [6] discovered that viral content on one platform can spill over to other platforms, leading to increased consumption on those platforms. Similarly, in advertising contexts, studies have shown that advertising in one medium can enhance the effectiveness of ads in another medium [7,8]. However, to the best of our knowledge, spillover effects in the context of influencers have not yet been examined.

On the other hand, emerging theories on influencer marketing have highlighted the importance of influencer characteristics, such as authenticity and credibility, in driving consumer engagement [4,9,10]. However, in the influencer literature, influencers have often been examined independently without considering their interdependencies. These interdependencies may be contingent on influencer popularity; however, this aspect has not yet been examined. Given the increasing advertising expenditure on influencers, it is both theoretically and managerially important to examine these popularity-based interdependencies to optimize influencer marketing strategies.

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To address these gaps, this study explores popularity-based interdependencies among multiple influencers recruited for the same marketing campaign. Specifically, we examine whether within-group spillover effects exist among micro- or macro-influencers, and whether cross-group spillover effects occur between these groups. We employ a median-split approach to categorize influencers into micro- and macro-influencers and examine their within- and cross-group spillovers during a marketing campaign by addressing the following research questions: (RQ1) What are the differential within-group spillover effects for micro-influencers compared to macro-influencers? (RQ2) What are the differential cross-group spillover effects between micro-influencers and macro-influencers?

To test our hypotheses regarding within- and cross-group spillover effects in influencer marketing strategies, we collected data from a real marketing campaign covering 163 promotional videos from 111 influencers over a 38-month period. The fixed-effects panel data regression indicates that both within- and cross-group spillover effects are positive and significant for micro-influencers only. Furthermore, to explore consumer perceptions that may contribute to the varying within- and cross-group spillover effects between micro- and macro-influencers, we conducted an online survey of 320 social media users. The independent sample *t*-test reveals that consumers trust micro-influencers more and perceive them as more authentic and credible than macro-influencers.

This study makes three key contributions to the existing literature. First, it advances research on spillover effects by examining these effects among online influencers in the context of short video-enabled influencer marketing [11]. Second, it augments the emerging influencer marketing literature by extending beyond the isolated effect of individual influencers and examining the differential within- and cross-group spillover effects for macro- versus micro-influencers [3,12]. Third, it enriches the influencer marketing and general social media marketing literature by investigating consumers' different perceptions of micro- versus macro-influencers. Specifically, we demonstrate that consumers trust micro-influencers more than macro-influencers. Our study provides important managerial insights into how firms can leverage the differential within- and cross-group spillover effects between micro- versus macro-influencers when planning cost-effective influencer marketing strategies.

2. Literature review

This study draws upon two key research streams: influencer marketing and spillover effects. The influencer marketing literature provides insights into the key factors driving the effectiveness of influencer marketing, such as influencer popularity. The literature on spillover effects explores the benefits of spillovers in various business contexts. By integrating these two streams, we develop a framework that leverages popularity-based spillover effects to enhance the overall effectiveness of influencer marketing strategies.

2.1. Influencer marketing

Social media influencers are individuals who experience fame through social media and build a follower group based on their established expertise [4,13]. In contrast to endorsements from traditional celebrities who gain fame independently of social media, consumers develop stronger feelings of intimacy and personal connection with social media influencers. This has led to the tremendous growth of influencer marketing within firms [14]. Numerous studies have examined diverse factors affecting the effectiveness of influencer marketing, including influencers' expertise [15]; activity [16]; follower size [12,17]; influencer–follower relationships [18]; and message content of sponsored posts [17,19,20].

Among the various factors driving the effectiveness of influencer marketing (see Table 1 for the most recent advertising studies on influencer marketing), the most readily observable factor for both practitioners and researchers is influencers' follower base size [12,17]. Regarding the effects of follower base size, studies have yielded contradictory findings [4,16,17,21]. On the one hand, some studies have documented a positive effect of a larger follower base. For instance, De Veirman [4] found that an influencer's follower base positively affects consumer attitudes toward the influencer and the products endorsed by the influencer, because consumers may ascribe opinion leadership to the influencer because of their popularity. Gong [17] found that influencers with more followers are more influential in attracting new followers to an endorsed brand. On the other hand, some researchers have identified a positive effect of a smaller follower base. Particularly, they have found that micro-influencers, compared to macro-influencers, develop a stronger sense of intimacy through frequent two-way communication with their followers [9,22,23]. Despite the valuable insights provided by these studies, influencers have often been examined in isolation, overlooking the spillover effects among influencers.

Table 1
Literature review on influencer marketing.

Paper	Theory perspective/ basis	Context	Method	Relevant finding
Scholz [24]	Practice theory	Beauty consumption	Ethnography	This paper explains how consumers incorporate influencer content into their own practice performances.
Yang, Chuenterawong, and Pugdeethosapol [25]	Source–message fit theory	Instagram posts on BLM	ANOVA and sentiment analysis	Audiences' reactions to content related to the BLM (The Black Lives Matter) movement differ by poster (i.e., black versus non-black influencers).
Brooks, Drenten, and Piskorski [26]	Celebrity Capital Life Cycle theory	Interviews with advertising industry practitioners and influencers	Qualitative study (Grounded theory approach)	The authors explain the process by which social media influencers acquire celebrity capital.
Rundin and Colliander [27]	NA as a theory development paper	21 fashion influencers on Instagram over three years	An inductive, qualitative netnography approach	The authors conceptualize three types of influencer roles in advertising collaborations, including spokesperson, cocreator, and co-owner.
Park et al. [9]	Endorser and brand authenticity theory	Social media influencer (SMI) advertising	Lab experiment	Advertising effectiveness from micro-influencers is stronger than that from macro-influencers because micro-influencers are perceived as more authentic.
Kim, Duffy, and Thorson [28]	Persuasion knowledge theory	Social media influencers in corporate reputation advertising	Lab experiment	Introducing a commercial from an influencer (versus the company itself) creates more positive message attitudes and corporate reputation via reduced persuasion knowledge.
Lou [29]	NA as a theory development paper	Interviews with 26 social media users who have a favorite influencer	Qualitative study	The author proposes a new concept, trans-parasocial relation, in comparison to traditional parasocial relation, which captures the co-created, reciprocal, and interactive relation between influencers and their followers.

2.2. Spillover effects

Spillover effects refer to the unintended impact of events in one setting on other events in a different, indirectly connected setting [6,30,31]. Researchers have examined spillover effects in various business contexts. For instance, in the context of online reviews, Kwark et al. [32] discovered that the ratings of substitute and complementary products have different spillover effects on the purchase of the focal product. Specifically, the mean rating of substitute products negatively affects the purchase of the focal product, whereas that of complementary products has a positive effect. In the context of viral content consumption, Krijestorac, Garg, and Mahajan [6] identified a spillover effect of viral content from a lag platform on the consumption of the same viral content on the lead platform that released the viral content.

In terms of advertising, existing studies have identified spillover effects among combinations of different advertising media. For instance, Havlena et al. [33] focused on three advertising media (TV, print, and online campaigns), revealing that a spillover effect exists when these three advertising sources are utilized simultaneously. Lee and Vakratsas [7] explored the asymmetric spillover effects of TV and online ads and found that prior exposure to online ads significantly increases the effectiveness of offline TV ads; however, prior TV ads do not catalyze the effectiveness of online ads. In the context of social media advertising, Unnava and Aravindakshan [8] explored the spillover effects arising from ads on multiple social media platforms, such as Facebook, Instagram, Twitter/X, and YouTube, and examined how ads on one platform generate consumer engagement with another. Nevertheless, to the best of our knowledge, spillover effects in influencer contexts have not yet been examined.

3. Theoretical framework: Spillover effects in influencer marketing

As brands increasingly recruit multiple influencers to simultaneously promote their products (e.g., within the same month), spillover effects among these influencers present a critical opportunity for optimizing an effective influencer portfolio. To assist brands in capitalizing on these spillover effects, this study investigates their operation within social media influencer marketing, specifically examining how sponsored content from influencers, both within the same popularity tier and across different tiers, affects the conversion performance of a focal influencer endorsing the same brand. We conceptualize these spillovers through two dimensions: within- and cross-group effects, tied to the relative popularity of the influencers involved. Within-group spillover effects occur when same-group ads—sponsored content posted by influencers of identical popularity tiers (e.g., multiple micro- or macro-influencers promoting the same brand)—influence the conversion performance of a focal influencer within that tier. For instance, when several micro-influencers endorse a product concurrently, their collective promotions may amplify or diminish the conversion effectiveness of a focal micro-influencer's campaign for that brand.

Conversely, cross-group spillover effects emerge from opposite-group ads, where influencers from divergent popularity tiers (e.g., macro- and micro-influencers) promote the same brand. The sponsored content of one group affects the focal influencer's performance in the other group. For example, a macro-influencer's endorsement could alter consumer responses to a micro-influencer's posts for the same product and vice versa. By delineating these dynamics, we specifically examine whether micro- and macro-influencers benefit from within- and cross-group spillover effects to different extents.

4. Research hypotheses

4.1. Within-Group spillover effects for Micro- versus macro-influencers

Despite having smaller follower bases, micro-influencers are perceived as more authentic and trustworthy because of their higher engagement levels and personalized interactions with followers [34]. When consumers are repeatedly exposed to ads from multiple micro-influencers, this perceived authenticity creates an advantage that leads them to be more likely to trust product endorsements, thereby increasing conversion rates [35].

In contrast, macro-influencers, who have larger audiences, often face the erosion of consumer trust because of their commercial motivations and lower engagement rates [4]. Consequently, repeated exposure to ads from multiple macro-influencers may backfire, as consumers increasingly attribute these endorsements to monetary incentives rather than genuine product advocacy, leading to lower conversion rates. These dynamics suggest a clear divergence in spillover effects between the two groups. Therefore, we hypothesize the following.

H1. The within-group spillover effect among micro-influencers is stronger than that among macro-influencers.

4.2. Cross-Group spillover effects for Micro- versus macro-influencers

Along with the within-group spillover effect, the cross-group spillover effect also varies with influencers' popularity. Macro-influencers, due to their broad reach and visibility, are often promoted by platform algorithms, exposing numerous consumers to their ads even if some consumers do not follow them [36]. This initial exposure creates product awareness, which is later reinforced by ads from micro-influencers. When consumers witness the same product endorsed by a micro-influencer they follow, the micro-influencer's perceived authenticity and trustworthiness increase conversion likelihood [37].

Conversely, when micro-influencers' ads are viewed first, their limited reach reduces the likelihood of generating widespread awareness. Subsequent exposure to macro-influencers' ads is less effective because of consumers' lower trust in their endorsements [35]. Therefore, we hypothesize the following.

H2. The cross-group spillover effect from macro-influencers to micro-influencers is stronger than that from micro-influencers to macro-influencers.

5. Data and methodology

5.1. Empirical context

Our dataset features a series of WOM campaigns on YouTube, conducted by a Chinese company promoting a mobile application targeting the Brazilian market. Before selecting influencers, the company thoroughly researched their performance metrics, including follower count, exposure, views, number of comments, and posting frequency. They also reviewed influencers' collaboration history, associated brands, influence, and marketing capabilities. Based on this comprehensive research, the company shortlisted influencers whose audience demographics matched the company's target users and whose content aligned with the brand's image. Given that the endorsed product is a video application featuring humorous and trendy lifestyle content, influencers known for their humor- and fashion-related expertise were considered ideal partners.

The company negotiated advertising fees with selected Brazilian influencers for sponsored content, basing payments on each influencer's popularity. Influencers were required to create sponsored videos, highlight the advantages of the mobile application, and upload the sponsored content to YouTube. Additionally, a link for viewers to download the mobile application was included in the introduction section of the sponsored videos, which helped the firm achieve its conversion goal for the WOM campaign.

Our final dataset comprised 163 promotional campaign videos from 111 local influencers over a 38-month period from April 2016 to May 2019. The dataset was constructed from two sources. First, we collected

a list of influencers collaborating with the firm and tracked their channels on YouTube. Specifically, we obtained the following information from each influencer's ad and channel: release date of the sponsored ad, number of channel subscriptions, and launch date of the channel. These data correspond to the variables of interest, namely the number of ad releases, influencer popularity, and influencer experience. Second, we tracked the monthly conversion performance (i.e., download rate) of each campaign video using Google Analytics.

5.2. Unit of analysis and measures

Our unit of analysis is individual video i (i= 1, …, 163) of influencer j (j= 1, …, 111) in month t (t= 1, …, 38). We median-split the influencers based on their subscription numbers (i.e., number of followers) and defined those in the more popular group as macro-influencers and those in the less popular group as micro-influencers [38,39]. To capture within- and cross-group spillover effects of ad repetition, we constructed *Same-Group Ad Release* ($RELEASESG_{it}$) and *Opposite-Group Ad Release* ($RELEASEOG_{it}$), respectively. *Same-Group Ad Release* ($RELEASESG_{it}$) represents the number of ads released by influencers in the same popularity group in the same month, excluding those released by the focal influencer in the same month. *Opposite-Group Ad Release* ($RELEASEOG_{it}$) denotes the number of ads released by influencers in the opposite-popularity group in the same month. Specifically, for the micro-influencer group, within-group spillover effects are measured by the effects of ads released by other micro-influencers in the same month on the focal influencer, while cross-group spillover effects are measured by the effects of ads released by all macro-influencers in the same month. However, for the macro-influencer group, within-group spillover effects are measured by the effects of ads released by other macro-influencers in the same month, while cross-group spillover effects are measured by the effects of ads released by all micro-influencers in the same month.

To control for the effect of previous ad releases on the conversion of current ads [8], we generated ad stock variables as the sum of past advertising effects, which are assumed to diminish at a particular rate (i.e., the carryover rate[1]). Specifically, we calculated the ad stock for ads within the same popularity group (*Same-Group Ad Stock,* abbreviated $STOCKSG_{it}$), ads from the opposite-popularity group (*Opposite-Group Ad Stock,* abbreviated $STOCKOG_{it}$), and ads from the same influencer (*Own Ad Stock,* abbreviated $STOCKOW_{it}$) using the following equations [40].

$$STOCKSG_{it} = STOCKSG_{i,t-1} * \theta + RELEASESG_{i,t-1} \quad (1)$$

$$STOCKOG_{it} = STOCKOG_{i,t-1} * \theta + RELEASEOG_{i,t-1} \quad (2)$$

$$STOCKOW_{it} = STOCKOW_{i,t-1} * \theta + RELEASEOW_{i,t-1} \quad (3)$$

where θ indicates the carryover rate.

As a few influencers posted more than one sponsored ad each month, we also controlled for the effect of the focal influencer's ads and included *Own Ad Release* ($RELEASEOW_{it}$), defined as the number of ads released by the focal influencer in the same month. As influencers' experience in operating media channels varies, we also included *Influencer Experience* ($CHANNELAGE_{jt}$), defined as the number of months since the launch of the influencer's YouTube channel. The dependent variable, *Download Rate (DOWNLOAD)*, is defined as the natural logarithm of the number of downloads of video i in month t.

We summarize the measurements in Table 2 and report the descriptive statistics in Table 3.

5.3. Model specification

As the data are longitudinal in nature, with repeated observations for each ad i (i= 1, …, 163) over each month t (t= 1, …, 38), a panel data analysis is needed. To capture ad-specific unobservable factors that may affect ad conversion performance, such as influencer popularity, tenure, and expertise, we included ad fixed effects (η_i) in the model. We divided the sample of influencers into two halves based on the median level of influencer popularity and tested the proposed hypotheses by estimating the following equation separately for the subsamples of macro- and micro-influencers.

$$DOWNLOAD_{it} = \beta_0 + \beta_1 RELEASESG_{it} + \beta_2 RELEASEOG_{it} + \beta_3 RELEASEOW_{it} + \beta_4 STOCKOW_{it} + \beta_5 STOCKSG_{it} + \beta_6 STOCKOG_{it} + \beta_7 CHANNELAGE_{jt} + \eta_i + \omega_{it} \quad (4)$$

6. Results

Table 4 presents the results of the fixed-effects panel data regression analysis. Models 1 and 2 present the results for macro- and micro-influencers, respectively.

Table 2
Measurement table.

Variable	Measurement
Downloads (DOWNLOAD)	The natural logarithm of the number of downloads from influencer i in month t
Same-group Ad Release (RELEASESG)	The number of ads released by influencers in the same popularity group in the same month, excluding the number of ads released by the focal influencer in the same month.
Opposite-group Ad Release (RELEASEOG)	The number of ads released by influencers in the opposite popularity group in the same month.
Own Ad Release (RELEASEOW)	The number of ads released by the focal influencer in the same month.
Same-group Ad Stock (STOCKSG)	$STOCKSG_{it}$= $STOCKSG_{i,t-1}$ * θ+ $RELEASESG_{i,t-1}$
Opposite-group Ad Stock (STOCKOG)	$STOCKOG_{it}$= $STOCKOG_{i,t-1}$ * θ+ $RELEASEOG_{i,t-1}$
Own Ad Stock (STOCKOW)	$STOCKOW_{it}$= $STOCKOW_{i,t-1}$ * θ+ $RELEASEOW_{i,t-1}$
Month of Channel (CHANNELAGE)	The number of months since the establishment of the channel

Notes:.
(1) The same- and opposite-popularity group is based on a median split on subscriptions.
(2) θ indicates the carryover rate, which is used to capture the ad stock effect in the regression model.

[1] To set the optimal carryover rate, we ran a set of experiments in which we changed the parameter values and compared the model fit indicated by the information criteria. In the experiments, the carryover rate θ was changed from 0 to 1 in steps of 0.05 (Liu et al. 2010). We used the information criteria (e.g., the AIC and BIC) to select the best models using maximum likelihood estimators. The ideal carryover rate is the rate that minimizes both the AIC and BIC. The AIC and BIC values were minimized at a carryover rate of 0.6 for both macro- and micro-influencer groups. A carryover rate of 0.6 indicates that an influencer's sponsored ad remains effective in the following months and that the effect diminishes by approximately 90% in 6 months.

Table 3
Descriptive statistics.

		1	2	3	4	5	6	7	8
1	Downloads[a]	1							
2	Month of Channel	−0.172*	1						
3	Same-group Ad Release	0.067*	−0.010	1					
4	Opposite-group Ad Release	0.080*	0.003*	0.489*	1				
5	Own Ad Release	0.271*	−0.110*	0.131*	0.123*	1			
6	Same-group Ad Stock	−0.190*	0.114*	0.206*	0.269*	−0.258*	1		
7	Opposite-group Ad Stock	−0.172*	0.135*	0.280*	0.208*	−0.246*	0.753*	1	
8	Own Ad Stock	0.196*	−0.178*	0.097*	0.121*	−0.103*	0.000	0.014	1
	Mean	5.338	51.418	2.276	2.327	0.088	5.397	5.530	0.214
	S.D.	1.776	23.481	2.374	2.456	0.354	3.689	3.822	0.409
	Min	0	6	0	0	0	0	0	0
	Max	10.645	136	9	9	5	15.597	15.592	5
	Median	5.288	48	1	1	0	4.667	4.760	0.280

Notes: (1)* $p < .05$;.
(2) [a] Natural log-transformed.

Table 4
Fixed-effect panel data regression on the natural logarithm of downloads.

	Subsample Analysis	
	(1)	(2)
	Macro-influencers	Micro-influencers
Same-group Ad Release	0.012	0.026**
	(1.062)	(1.963)
Opposite-group Ad Release	0.000	0.032**
	(0.019)	(2.161)
Month of Channel	−1.189***	−1.290***
	(−30.611)	(−33.633)
Own Ad Release	0.236***	0.171***
	(23.133)	(12.709)
Own Ad Stock (STOCKOW)	0.127***	0.113***
	(11.733)	(7.605)
Same-group Ad Stock	0.008	−0.050**
	(0.569)	(−2.401)
Opposite-group Ad Stock	−0.069***	−0.011
	(−4.230)	(−0.581)
Ad Fixed Effect	Yes	Yes
Observations	1531	1363
Within-group R Square	0.699	0.667

Notes: The coefficients are standardized beta coefficients; t-statistics are in parentheses.
* $p < .10$. ** $p < .05$. *** $p < .01$ (all two-tailed).

6.1. Within- and cross-group spillover effects

Same-Group Ad Release ($RELEASESG_{it}$) is insignificant and positive for macro-influencers ($\beta = 0.012$, *n.s.*); however, it is significant and positive for micro-influencers ($\beta = 0.026$, $p < .05$). This indicates that within-group spillover effects exist only for micro-influencers; hence, H1 is supported. *Opposite-Group Ad Release* ($RELEASEOG_{it}$) is insignificant and positive for macro-influencers ($\beta = 0.000$, *n.s.*); however, it is significant and positive for micro-influencers ($\beta = 0.032$, $p < .05$). This suggests that cross-group spillover effects exist only for micro-influencers; hence, H2 is supported. Overall, micro-influencers enjoy both within-group spillover effects from micro-influencer peers and cross-group spillover effects from macro-influencers, whereas macro-influencers benefit neither from within-group spillover effects from macro-influencer peers nor cross-group spillover effects from micro-influencers.

6.2. Effects of control variables

For the macro-influencer group, a positive and significant effect was observed from the focal influencer's own ad stock (*Own Ad Stock*) ($\beta = 0.127$, $p < .01$), an insignificant effect from the same-group ad stock (*Same-group Ad Stock*) ($\beta = 0.008$, *n.s.*), and a negative and significant effect from the opposite-group ad stock (*Opposite-group Ad Stock*) ($\beta = -0.069$, $p < .01$). For the micro-influencer group, a positive and significant effect from the focal influencer's own ad stock exists (*Own Ad Stock*) ($\beta = 0.113$, $p < .01$), a significant and negative effect from the same-group ad stock (*Same-group Ad Stock*) ($\beta = -0.050$, $p < .05$), and an insignificant effect from the opposite-group ad stock (*Opposite-group Ad Stock*) ($\beta = -0.011$, *n.s.*). These two sets of results indicate that prior micro-influencer ads (released within six months) dilute the effectiveness of newly released macro- and micro-influencer ads. Regarding other control variables, influencer experience (*Channel Age*) presents a significantly negative effect on conversion for both macro-influencers ($\beta = -1.189$, $p < .01$) and micro-influencers ($\beta = -1.290$, $p < .01$). The number of ads from the focal influencer (*Own Ad Release*) presents a significantly positive effect on conversion for both macro-influencers ($\beta = 0.236$, $p < .01$) and micro-influencers ($\beta = 0.171$, $p < .01$).

6.3. Additional survey: consumer perceptions of Micro- versus macro-influencers

To examine consumers' perceived authenticity, credibility, and trust in macro- and micro-influencers, we conducted an online survey to complement the field data. Our survey targeted consumers aware of influencer marketing by inquiring about their experiences of encountering sponsored content from influencers on social media platforms. By measuring consumers' perceptions toward influencers, we examined whether a significant difference exists in consumers' perceived authenticity, credibility, and trust of macro- and micro-influencers.

When designing the survey questionnaire, we adopted measurements of authenticity, credibility, and trust previously validated in celebrity endorsement research in the influencer marketing context (see Appendix A for the measurement items). Apart from the three key constructs, the survey also included questions screening respondents' experience with sponsored content, influencer and respondent characteristics (i.e., nicknames, social media platforms, and follower base size), and respondents' demographics.

In the survey, we first introduced the concept of content sponsored by social media influencers and asked respondents about their experiences with such content. Next, we inquired about their social media usage and requested that they recall a specific social media influencer they follow, along with the number of followers that the influencer has. We then asked the respondents to assess their perceptions of the influencer's trustworthiness, authenticity, and credibility.

We collected the data in July 2022 by administering the survey using Amazon Mechanical Turk (MTurk). MTurk is an online survey platform on which individuals volunteer to participate in research surveys in exchange for monetary incentives [41]. It has become a significant resource for data collection in business research, as it enables researchers to recruit more demographically diverse populations than just

college students.

To ensure the relevance of the target participants, strict selection criteria were applied. Respondents were required to have prior exposure to product endorsements from social media influencers on leading social media platforms. Given that the respondents were English-speaking, they were required to use at least one of the following major English-language social media platforms: Facebook, LinkedIn, Instagram, Reddit, Pinterest, Snapchat, TikTok, Twitch, Twitter/X, or YouTube. Each participant was compensated with $2 for completing the survey. This compensation was competitively priced compared to similar studies on MTurk, reducing the risk of inattentive responses. After excluding 85 respondents who had no experience with influencer marketing, we obtained 515 valid responses covering influencers on various social media platforms.

Given the possibility that respondents may inaccurately recall an influencer's follower count, we verified this information using the aforementioned social media platforms. Based on the verified follower counts, we classified the influencers reported by survey respondents into two subsamples: 210 macro-influencers and 305 micro-influencers[2]. Based on a two-sample *t*-test, we found that consumers perceive micro-influencers as more authentic and credible than macro-influencers and trust micro-influencers more than macro-influencers. Fig. 1 presents the mean levels of perceived authenticity, credibility, and trust of macro- and micro-influencers. The average authenticity scores for the macro- (5.69) and micro-influencer subgroup (5.81) vary significantly ($t = -1.53$, $p < .10$). The results are consistent for the average credibility score (macro-influencer subgroup, 5.63; micro-influencer subgroup, 5.77; significant difference: $t = -1.83$, $p < .05$), as well as average trust score (macro-influencer subgroup, 5.31; micro-influencer subgroup, 5.50; significant difference, $t = -2.07$, $p < .05$).

6.4. Robustness check

To test the generalizability of our main results on the spillover effects between micro- and macro-influencers, we conduct additional analyses to examine the spillover effects between more and less popular influencers on the focal influencer's conversion performance. Instead of using a median split on our dataset, we employ a continuous variable to measure the average popularity distance between the focal influencer and other more (or less) popular influencers. We also include the number of ads from more (or less) popular influencers as control variables. Specifically, we examine the interaction between the subscription distance with more (or less) popular influencers and the number of ads simultaneously observed from these influencers (see Table 5 for the stepwise results from the fixed-effect panel data regression).

We find a positive and significant interaction between the subscription distance with more popular influencers and the number of simultaneous ads from these influencers ($\beta = 0.304$, $p < .05$ in Model 4). Conversely, the interaction between the subscription distance with less popular influencers and the number of simultaneous ads from these influencers was negative but insignificant ($\beta = -0.068$, *n.s.*).

These results suggest that influencers benefit from positive spillover effects when associated with more popular influencers, but not with less popular ones. This is consistent with our initial findings using the median-split approach, where micro-influencers experienced positive spillover from macro-influencers, while macro-influencers did not benefit from spillover from micro-influencers.

[2] Based on the survey data collected, we employed the industry criteria (i.e., 1 million followers) to classify influencers into micro- and macro-influencers. The results show significant difference in terms of authenticity, trust, and credibility. We also used a median-split based on influencers' follower base size; the results show significant difference in trust.

7. Discussion

Despite substantial investment in influencer marketing and significant cost discrepancies between micro- and macro-influencers, the potential interactions among micro- versus macro-influencer peers, as well as between micro- and macro-influencers, have not been adequately explored by marketing researchers. In this study, we categorize micro- and macro-influencers into two distinct groups and examine their within- and cross-group spillovers during a marketing campaign. Utilizing a fixed-effects panel data regression on our dataset, which comprises 163 sponsored video ads from 111 local influencers over a 38-month period, we found that micro-influencers achieve higher conversion rates when more micro- and macro-influencers release ads simultaneously. In contrast, macro-influencers do not benefit from the simultaneous release of ads by either micro-influencers or macro-influencers.

7.1. Theoretical contributions

Our findings offer three important theoretical contributions to the literature. First, we extend the concept of the spillover effect to the context of influencer marketing. Given the theoretical significance of influencer popularity [4], our study enriches the spillover literature by specifying the differential spillover effects among more and less popular influencers in the context of influencer marketing.

Second, we contribute to the emerging influencer marketing and general social media marketing literature by extending beyond the individual features of influencers (e.g., popularity) and focusing on their interactions as a collective [3,12]. Specifically, because existing studies on the effectiveness of influencer marketing strategies have primarily focused on the isolated effects of influencers' *individual* features, such as personality traits, sociodemographic backgrounds, and knowledge or expertise [42], our study underscores the theoretical significance of the interdependencies among multiple influencers involved in a long-term advertising campaign.

Third, we integrate the literature on variations in advertising repetition into the context of influencer marketing. We propose that influencers participating in the same advertising campaign can deliver a consistent message about a product while crafting uniquely sponsored content based on their expertise. This approach ensures that consumers receive both consistent and engaging information, thereby maintaining their interest. We theoretically highlight the advantage of influencer marketing as a unique form of ad repetition variation strategy, as influencers' ads are executed in unique ways based on their expertise, offering fresh value to consumers [10,43].

7.2. Managerial implications

Our findings have several important implications for brand managers employing influencers in social media marketing campaigns, as well as for platform managers. For brand managers, optimizing the composition of their influencer portfolios is crucial. Given the substantial cost discrepancy between micro- and macro-influencers, brand managers should consider a balanced approach when selecting an effective influencer portfolio. Thus, an overreliance on costly macro-influencers should be avoided. Instead, brand managers should leverage micro-influencers' cost-effectiveness. For instance, a firm could allocate a larger percentage of its budget to micro-influencers who, despite their smaller follower bases, can collectively generate significant engagement and conversion. This approach can reduce overall advertising expenses while maintaining effectiveness.

Additionally, brand managers should adopt a holistic view when attributing conversions, considering the collective impact of influencers rather than evaluating them individually. Recognizing the synergistic effects within and between groups of influencers is essential. Despite their smaller reach, micro-influencers can have significant collective

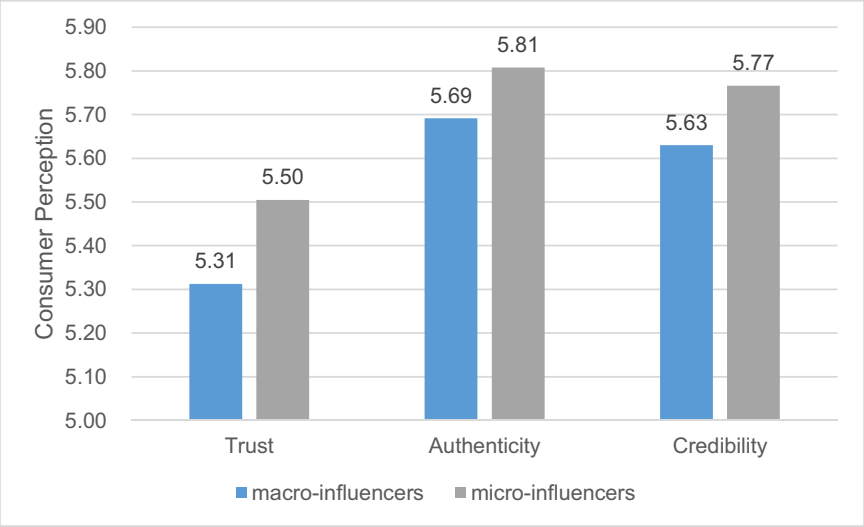


Fig. 1. Perceived trust, authenticity, and credibility for macro and micro-influencers.

Table 5
Robust check using popularity distance with more (less) popular influencers.

	(1)	(2)	(3)	(4)
Own Ad Release	0.179***	0.179***	0.179***	0.179***
	(25.580)	(25.515)	(25.557)	(25.473)
Month of Channel	−1.958***	−1.938***	−1.959***	−1.937***
	(−29.403)	(−28.765)	(−29.404)	(−28.733)
Number of Ads from More Popular Influencers (*Ads Higher*)	0.188***	−0.105	0.183***	−0.145
	(6.169)	(−0.677)	(5.836)	(−0.910)
Popularity Distance with More Popular influencers (*Distance Higher*)	0.034**	0.029**	0.029*	0.021
	(2.360)	(1.984)	(1.811)	(1.243)
Number of Ads from Less Popular Influencers (*Ads Lower*)	0.329***	0.324***	0.351***	0.359***
	(10.666)	(10.483)	(7.762)	(7.924)
Popularity Distance with Less Popular influencers (*Distance Lower*)	0.891***	0.952***	0.907***	0.984***
	(4.068)	(4.303)	(4.115)	(4.407)
*Distance Higher * Ads Higher*		0.274*		0.304**
		(1.927)		(2.098)
*Distance Lower * Ads Lower*			−0.042	−0.068
			(−0.662)	(−1.060)
Ad Fixed Effect	Yes	Yes	Yes	Yes
Observations	3134	3134	3134	3134
Within-group R Square	0.697	0.698	0.697	0.698

Notes: The coefficients are standardized beta coefficients; t-statistics are in parentheses.
* $p < .10$. ** $p < .05$. *** $p < .01$ (all two-tailed).

power, particularly when supported by macro-influencers, who drive initial awareness. Instead of attributing conversion performance solely to individual influencers, managers should assess the roles of both micro- and macro-influencers throughout the consumer decision journey from awareness to purchase. For instance, a brand may track the initial awareness generated by macro-influencers and subsequent conversions driven by micro-influencers, ensuring that both types of influencers are appropriately credited for their contributions.

For influencers, understanding their role within the broader marketing strategy is key. Micro-influencers should recognize their collective power and the value they bring to campaigns, even if their individual follower bases are smaller. By building strong and authentic relationships with audiences, micro-influencers can significantly impact campaign success. Collaborating with other micro-influencers can further amplify their reach and effectiveness. On the other hand, macro-influencers should be aware of their role in driving initial awareness and traffic spillovers to micro-influencers. They should strive to create high-quality content that resonates with a wide audience, setting the stage for micro-influencers to drive conversions. Both micro- and macro-influencers should seek opportunities for collaboration and leverage each other's strengths to maximize campaign impact.

For platform managers, enhancing platform policies and tools to support the effective use of both micro- and macro-influencers is important. Platform managers should develop features that facilitate influencer identification and collaboration across tiers to ensure that brand managers can easily create balanced and effective influencer portfolios. For instance, platforms could introduce advanced analytic tools that highlight the synergistic effects of influencer collaboration, assisting marketers in optimizing their strategies. A platform may offer a dashboard that visualizes the combined impact of micro- and macro-influencers on campaign performance, thus aiding marketers in making informed decisions.

7.3. Limitations and future research

Our study is limited to a few dimensions, which present opportunities for future research. First, we collected field data at the video level and examined the interactions among influencers. Our results reveal within-group spillover effects among micro-influencers' peers and between-group spillovers from macro-influencers to micro-influencers. However, we were unable to empirically validate the underlying mechanisms at the viewer level. Our online survey indicates that consumers perceive micro-influencers as more authentic than macro-influencers, suggesting a possible mechanism for these effects. Future studies should conduct laboratory experiments to test the underlying mechanism of this spillover effect.

Second, in our field data, we measured key variables on a monthly basis. For example, we calculated the within-group spillover for micro-influencers as the number of ads released in the same popularity group within the same month. However, we did not consider the sequence of ad releases within the same month. For instance, when hypothesizing a differential effect in the spillover effect, we posited that viewers may gain awareness of the product during their initial viewing of a macro-influencer's video and subsequently decide to download the product upon viewing a video from a micro-influencer. These sequential responses could not be empirically tested using our field or survey data.

Hence, future research should consider different ad release sequences and empirically test the exact consumer decision journey when exposed to multiple influencer ads in a refined laboratory experiment.

Third, our dataset is limited to one product and company, which restricts the generalizability of our findings. Future research could enhance the generalizability of these results by utilizing a more diverse dataset that includes multiple product categories and companies. This would facilitate the examination of how the effectiveness of influencer marketing varies across different product categories and company characteristics.

CRediT authorship contribution statement

Li Ji: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Xingyu Chen:** Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Ling Jiang:** Writing – original draft, Validation, Methodology, Conceptualization. **Justin T. Huang:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper. No financial, personal, or professional affiliations have influenced the research and findings presented in this manuscript. All authors have contributed to the work and have approved the final version of the manuscript for submission.

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[1]Appendix A. Measurement items for key variables in the survey

Variable	Measurement	Reference
Authenticity	(1) This influencer tries to act in a manner that is consistent with her held values, even if others criticize or reject her for doing so (2) This influencer cares about openness and honesty in close relationships with others (3) In general, this influencer places a good deal of importance on others understanding who she truly is (4) People can count on this influencer being who she is, regardless of the situation	[34]
Credibility	(1) The influencer is convincing (2) The influencer is believable (3) The influencer is unbiased	MacKenzie and Lutz (1989)
Trust	(1) I trust this influencer (2) I rely on this influencer (3) This is an honest influencer (4) This influencer is safe	Reinikainen et al. (2020)

Appendix B. Demographic statistics of the survey sample

Variables	Type	Frequency	Percentage (%)
Gender	Female	214	41.55
	Male	278	53.98
	Non-binary	1	0.19
	Prefer not to answer	22	2.14
Age	18–35	259	50.29
	36–55	203	39.42
	56 and above	35	6.80
	Prefer not to answer	18	3.50
Income	<60,000 USD	289	56.12
	60,001–100,000 USD	148	28.74
	100,001–140,000 USD	31	6.02
	>140,000 USD	16	3.11
	Prefer not to answer	31	6.02
Time Spent on Social Media (Daily)	Less than one hour	114	22.14
	One to three hours	280	54.37
	Three to five hours	85	16.50
	More than five hours	36	6.99
Social media platform used*	Facebook	30	5.83
	Instagram	193	37.48
	TikTok	1	0.19
	Twitch	31	6.02
	Twitter/X	39	7.57
	YouTube	221	42.91

Notes: (1)* Respondents may use more than one social media platform. The social media platform here refers to the one that the respondents use to follow the influencer whom they recall in the survey.

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